AYUSH WAGHE

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SKILLS

- **Languages:** Java, SQL
- **Backend:** Spring Boot, Spring MVC, Spring Security, Spring AI, Hibernate, JPA, REST APIs, Kafka
- **Databases & Data:** PostgreSQL, Qdrant
- **Engineering Tools:** Docker, Maven, Git, GitHub Actions, JUnit, Micrometer, Actuator, Postman
- **Frontend:** React, HTML/CSS
- **CS Fundamentals:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
- **Applied AI:** RAG, Hybrid Retrieval (BM25 + Vector Search), LLM Integration

EXPERIENCE

Tata Consultancy Services **July 2025 – Present**
Systems Engineer (Digital Candidate) **Mumbai**

- Architected and enhanced a modular Java framework supporting **~20K test cases** per module execution, improving reusability,scalability and maintainability through a structured component design.
- Optimized framework execution by implementing configurable batch-based Excel I/O (default batch size: 50), reducing number of I/O file operations and decreasing overall test execution time by **approximately 40%.**
- Abstracted test operations into reusable action and validation components with clear separation of concerns, reducing new test-case implementation to only locator configuration and action/validation mapping thus cutting development time for ~**100 test cases by 55%.**
- Improved framework reliability by implementing **configurable retry mechanism** for test failures and Excel data-integrity safeguards, including **pre-execution file-lock** detection and **rotating backups** for corruption prevention and recovery.
- Developed **GitHub Actions workflows** to automate Maven builds and module-level test execution, with configurable test parameters and automated collection of test reports and execution artifacts.

Samsung Research Institute (Bangalore) **August – December 2023**
Research Intern

- Engineered a TensorFlow-based computer vision solution for detecting ghost artifacts in HDR images as part of Samsung PRISM.
- Evaluated and optimized the detection pipeline to improve reliability across diverse HDR image datasets.

PROJECTS

Savings Saga | Java Backend Application (Spring Boot, React, PostgreSQL, Kafka, Docker) | [GitHub] | [Live demo]

- Built a **personal finance management application** implementing **15+ REST** endpoints for tracking transactions, budgets, recurring bills, spending analytics, and financial goals.
- Implemented secure stateless authentication, securing protected APIs using **JWT, HttpOnly cookies**, and **Spring Security**
- Designed **event-driven** bill reminders using **Apache Kafka**, and automated recurring-bill generation and overdue-bill processing with Spring scheduled jobs.
- Improved data-access efficiency using **PostgreSQL indexes**, paginated/sorted transaction APIs, JOIN FETCH to avoid **N+1 queries**, and implemented **Caffeine application-level caching** for frequently accessed user profiles and category metadata to reduce redundant database queries.

FinGuru | AI-Enabled Financial Platform (Spring AI, Spring Boot, PostgreSQL, Qdrant, RAG) | [GitHub] | [Live demo]

- Built a **Spring AI**-powered financial assistant with **tool calling** capability for transaction analysis and bill-management actions, plus **RAG-based** document Q&A backed by **PostgreSQL** and **Qdrant**.
- Designed a production-oriented **RAG service** using **hybrid BM25 and vector retrieval**, Qdrant, metadata filtering, reranking, and parallel asynchronous retrieval; reduced query latency by approximately **40%**.
- Developed LLM-driven document ingestion APIs and evaluation pipeline, dynamically selecting PAGE, DOCUMENT, or SEMANTIC chunking strategies and benchmarking retrieval using **Recall@K, Precision@K, and NDCG@K**, alongside LLM-based evaluation of relevance, faithfulness, and answer quality.
- Added operational safeguards including model-usage/credit controls, document and token limits, input validation, centralized exception handling, and **Micrometer/Actuator** metrics for LLM and retrieval latency.

CERTIFICATIONS

- AWS Certified Cloud Practitioner

ACHIEVEMENTS

- **Research publication:** Published a peer-reviewed research paper in **Scientific Reports (Nature Portfolio)** on machine learning techniques for banana plant health analysis. [Link]
- **Co-inventor on a published Indian patent application**: "Real-time HDR Ghost Artifact Detection and Inpainting for Safety Systems" (Application No. 202541118358).

EDUCATION

Vellore Institute of Technology ,Vellore (2025)
Computer Science and Engineering **CGPA: 8.7**